

Notes: Rajan and Zingales (AER, 1998)

Financial Dependence and Growth

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Industry-country panel and unit of observation	4
2.2	Country-level financial development and controls	4
2.3	Industry-level external dependence	5
2.4	Key descriptive objects	5
2.5	Timing and interpretation	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	The finance-growth mechanism	6
3.2	Basic empirical specification	6
3.3	Construction of external dependence	7
3.4	Other financing and investment measures	7
3.5	Weighted dependence as a country-level validation exercise	7
3.6	Growth decomposition	7
3.7	Cash flow versus investment intensity specification	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	What fails in standard cross-country growth regressions	8
4.2	Why the interaction design is informative	8
4.3	Why the U.S. dependence measure is plausible	9
4.4	Instruments for financial development	9
4.5	Robustness against omitted variables and reverse causality	9
4.6	External-finance channel versus general cost of capital	10
5	Tables and figures	10

5.1	Figure 1 and Table 1: life cycle of financing and industry rankings	10
5.2	Table 2 and Table 3: country finance measures and summary statistics	11
5.3	Table 4: baseline interaction results	11
5.4	Table 5 and Table 6: actual growth patterns and young-firm dependence	12
5.5	Table 7 and Table 8: alternative dependence measures and decomposition of growth	14
5.6	Table 9 and Table 10: omitted variables, reverse causality, and the cost-of-capital channel	15
6	Short summary	15
7	Conclusion	16
7.1	Core conclusion (what the paper establishes)	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** Does financial development foster economic growth by reducing firms' cost of raising **external finance**?
- **Setting.** The paper studies a cross-country, cross-industry panel of manufacturing industries over **1980–1990**. It combines country-level measures of financial development with industry-level measures of technological dependence on outside finance, constructed from U.S. firms.
- **Core challenge.** Standard finance-and-growth regressions are hard to interpret causally. Financial development may be correlated with omitted country traits, may simply anticipate future growth, or may itself respond to the financing needs of fast-growing industries.
- **Main methodological contribution.** The paper proposes a **within-country, between-industry interaction design**: compare whether industries that are more dependent on external finance grow disproportionately faster in countries with more developed financial markets. Country and industry fixed effects absorb broad country- and industry-level confounds.
- **Why this matters.** This design looks for a specific **channel** through which finance affects growth. It is therefore closer to a “smoking gun” for causality than standard cross-country correlations.
- **Main empirical question.** Do industries that need more external finance grow faster in countries with better financial systems?
- **Main empirical results.**
 - The interaction between industry external dependence and country financial development is **positive and statistically significant** across multiple measures of financial development.
 - Using total capitalization, the implied annual real growth differential is about **1.3 percentage points**; using domestic credit it is about **1.1**; using accounting standards it is about **0.9**; and with instrumented accounting standards it is about **1.0**.
 - The effect is especially strong for the **growth in the number of establishments**, suggesting that financial development disproportionately helps the formation of new firms.
 - The findings are robust to measuring external dependence with **young firms, mature firms, 1970s U.S. data, and Canadian data**.
 - The results survive controls for **human capital, per capita income, industry size, reverse-causality concerns**, and alternative decompositions using **cash flow and investment intensity**.
- **Economic takeaway.** Financial development matters for growth because it relaxes financing constraints in sectors that technologically need outside funds. It may also promote growth by disproportionately helping **new establishments** and therefore the emergence of new ideas and industries.

2 Data and measurement (key objects)

2.1 Industry-country panel and unit of observation

- **Industry data.** Value added and gross fixed capital formation come from the *Industrial Statistics Yearbook* of the United Nations Statistical Division.
- **Unit of observation.** The basic observation is an **industry-country cell**, with industries classified in ISIC manufacturing categories.
- **Sample restriction.** The analysis is restricted to **manufacturing industries**, in order to reduce the role of country-specific natural-resource endowments.
- **Time period.** Growth is measured over **1980–1990**.
- **Baseline sample size.** The basic regression with total capitalization uses **1,217** industry-country observations.
- **Dependent variable.** The main outcome is the **annual compounded growth rate in real value added** for an industry in a country over 1980–1990.

2.2 Country-level financial development and controls

- **Country sample.** The paper uses **41 countries**, drawn from the IFC's *Emerging Stock Markets Factbook* and matched with IMF and U.N. sources.
- **Main measures of financial development.**
 - **Total capitalization over GDP:** domestic credit plus stock market capitalization relative to GDP.
 - **Domestic credit to the private sector over GDP.**
 - **Accounting standards:** a disclosure-quality index from the Center for International Financial Analysis and Research.
 - **Accounting standards in 1983:** used as a predetermined alternative.
- **Why accounting standards matter.** The paper argues that accounting quality captures how well outsiders can evaluate firms, and hence how easily firms can raise external finance.
- **Other country controls.**
 - **Per capita income** from IMF data.
 - **Human capital:** average years of schooling in the population above age 25 from Barro–Lee.
- **Summary statistics.**
 - Mean annual real industry growth is about **3.4%**.
 - Mean total capitalization/GDP is **0.738**.
 - Mean domestic credit to the private sector/GDP is **0.377**.
 - Mean accounting standards score is about **61.3**.

2.3 Industry-level external dependence

- **Main idea.** External dependence is meant to proxy for an industry's **technological need for outside finance**, not the realized equilibrium use of finance in less developed countries.
- **Why U.S. firms?** The paper uses large U.S. listed firms because capital-market frictions are relatively small, so actual external financing is more likely to reflect **demand** rather than constrained supply.
- **Industry aggregation.** Firm-level external dependence is computed first, then aggregated to the **industry median** to avoid large firms dominating the measure.
- **Alternative dependence measures.**
 - all firms in the 1980s,
 - **mature firms** (listed more than 10 years),
 - **young firms** (listed fewer than 10 years),
 - U.S. firms in the **1970s**,
 - Canadian firms in the **1980s**.

2.4 Key descriptive objects

- **Most externally dependent industries in U.S. data.**
 - Drugs: **1.49**
 - Plastic products: **1.14**
 - Office and computing: **1.06**
 - Radio: **1.04**
 - Professional goods: **0.96**
- **Least externally dependent industries.**
 - Tobacco: **-0.45**
 - Pottery: **-0.15**
 - Leather: **-0.14**
 - Spinning: **-0.09**
 - Footwear: **-0.08**
- **Life-cycle fact.** Young firms raise a lot of external finance around the IPO year, continue to rely on it for roughly the first **10 years**, and then external finance fluctuates around zero. Net equity issuance is concentrated very early in firm life.
- **Cross-correlations.**
 - External dependence is strongly positively correlated with **investment intensity** (≈ 0.81).
 - It is strongly negatively correlated with **cash flow generation** (≈ -0.91).
 - The U.S. and Canadian dependence measures are highly correlated (≈ 0.77).

2.5 Timing and interpretation

- **Predetermined development.** The paper treats country financial development as predetermined with respect to industry growth over the 1980s.
- **Cross-industry design.** The identifying variation does not come from whether country k grows faster on average, but from whether **industry j** grows relatively faster inside country k when that industry is more financially dependent.
- **Interpretive caution.** The paper is not claiming that finance creates profitable opportunities from nothing; rather, finance helps firms pursue growth opportunities that require outside funding.

3 Models and empirical specifications (all equations + interpretation)

3.1 The finance-growth mechanism

The paper's core theoretical idea is that financial development reduces the wedge between the cost of internal and external funds. Industries that need external finance more intensely should therefore benefit more from a better financial system.

Interpretation. This is the channel the paper wants to identify directly, rather than relying on broad cross-country correlations between finance and growth.

3.2 Basic empirical specification

The main estimating equation is

$$\text{Growth}_{jk} = \alpha + \mu_j + \lambda_k + \beta \text{Share}_{jk,1980} + \gamma (\text{External Dependence}_j \times \text{Financial Development}_k) + \varepsilon_{jk}.$$

- Growth_{jk} : annual compounded real growth rate of value added in industry j , country k , over 1980–1990.
- μ_j : industry fixed effects.
- λ_k : country fixed effects.
- $\text{Share}_{jk,1980}$: industry j 's share in country k 's manufacturing value added in 1980.
- $\text{External Dependence}_j \times \text{Financial Development}_k$: the key interaction term.

Interpretation. Country fixed effects absorb all country-wide growth determinants, and industry fixed effects absorb all global industry-wide growth determinants. The coefficient γ therefore asks whether **financially dependent industries grow relatively faster inside financially developed countries.**

3.3 Construction of external dependence

At the firm level, external dependence is defined as

$$ED_i = \frac{\text{Capital Expenditures}_i - \text{Cash Flow from Operations}_i}{\text{Capital Expenditures}_i}.$$

Cash flow from operations is broadly defined as operating cash flow plus decreases in inventories, plus decreases in receivables, plus increases in payables.

The industry-level measure is the median of ED_i across U.S. firms in that industry.

Interpretation. A high value means a typical firm in the industry needs outside funds to finance investment. A negative value means internal cash flow more than covers capital expenditure.

3.4 Other financing and investment measures

The paper also defines:

$$\text{External Equity Finance}_i = \frac{\text{Net Equity Issues}_i}{\text{Capital Expenditures}_i},$$

$$\text{Investment Intensity}_i = \frac{\text{Capital Expenditures}_i}{\text{Net Property, Plant, and Equipment}_i},$$

$$\text{Cash Flow Intensity}_i = \frac{\text{Cash Flow from Operations}_i}{\text{Net Property, Plant, and Equipment}_i}.$$

Interpretation. These measures are used to check whether the core results are really about **external finance**, rather than simply capturing investment intensity or the general cost of capital.

3.5 Weighted dependence as a country-level validation exercise

To check whether the U.S.-based dependence measure has bite outside the United States, the paper computes a country's weighted average dependence:

$$\overline{ED}_k = \sum_j ED_j \cdot \text{Share}_{jk,1980}.$$

Interpretation. If industries that are externally dependent in the U.S. are also the ones that need outside finance elsewhere, then countries specializing more in these industries should exhibit larger financial systems. The paper finds a strong positive relationship between weighted dependence and capitalization.

3.6 Growth decomposition

Industry growth is decomposed into growth in the number of establishments and growth in average establishment size:

$$g_{jk}^{\text{number}} = \log N_{jk,1990} - \log N_{jk,1980},$$

$$g_{jk}^{\text{size}} = \log\left(\frac{VA_{jk,1990}}{N_{jk,1990}}\right) - \log\left(\frac{VA_{jk,1980}}{N_{jk,1980}}\right).$$

Interpretation. This allows the paper to ask whether financial development mainly helps existing firms expand, or whether it especially helps **new establishments** appear.

3.7 Cash flow versus investment intensity specification

To distinguish the “cost of external finance” channel from a more general “cost of capital” channel, the paper estimates specifications of the form

$$\begin{aligned} \text{Growth}_{jk} = & \dots + \phi_1(\text{Cash Flow Intensity}_j \times \text{Financial Development}_k) \\ & + \phi_2(\text{Investment Intensity}_j \times \text{Financial Development}_k) + \varepsilon_{jk}. \end{aligned}$$

Interpretation. If only the general cost of capital mattered, then industries with lots of internal cash should not be differentially affected by financial development. If instead finance reduces the wedge between internal and external funds, internal-cash-rich industries should behave differently from external-finance-dependent industries.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in standard cross-country growth regressions

- A simple positive correlation between finance and growth does not establish causality.
- Financial development may proxy for omitted variables, such as savings propensity, institutions, or broader development.
- Finance might simply anticipate future growth rather than cause it.
- Cross-country regressions face severe multicollinearity and few observations.

This paper’s response. Use interaction effects inside a country-industry panel, rather than country-level averages alone.

4.2 Why the interaction design is informative

- Country fixed effects purge country-wide growth determinants.
- Industry fixed effects purge global industry growth patterns.
- The key variation is whether industries that need more outside finance do **relatively better** in financially developed countries.

Interpretation. The identifying assumption is weaker than in a standard cross-country regression because the paper is comparing **differences across industries within the same country**.

4.3 Why the U.S. dependence measure is plausible

- U.S. listed firms are likely less financially constrained, so realized external financing better reflects desired financing.
- The relevant technological determinants of financing need—project scale, gestation period, and cash-flow timing—are assumed to be persistent across countries.
- The life-cycle pattern of financing supports the idea that many financing needs are structural and especially important early in a firm’s life.
- The results are robust to using old firms, young firms, 1970s data, and Canadian firms.

Interpretation. Even if the U.S. proxy is noisy, that noise should generally bias the estimated interaction **toward zero**, not create a spurious positive effect.

4.4 Instruments for financial development

The preferred financial-development measure is **accounting standards**, instrumented with predetermined institutional variables:

- indicators for legal origin (**British, French, German, Scandinavian**),
- an index of the **efficiency and integrity of the legal system**.

Interpretation. These instruments are meant to move financial development through institutional structure rather than through contemporaneous industry growth.

4.5 Robustness against omitted variables and reverse causality

The paper checks several threats:

- **Human capital proxy.** Adding external dependence $\times$ schooling barely changes the main interaction; the human-capital interaction is small and insignificant.
- **Economic development proxy.** Adding external dependence $\times$ log per capita income leaves the finance interaction positive and significant.
- **Reverse causality.** The results remain when focusing on industries that are initially small in a country and therefore unlikely to have shaped that country’s financial development.
- **Initial specialization.** Restricting to industries already large in a country also leaves the coefficient essentially unchanged.

Interpretation. These checks make it hard to explain the results purely by omitted comparative advantage or by finance developing in response to the same industries whose later growth the paper studies.

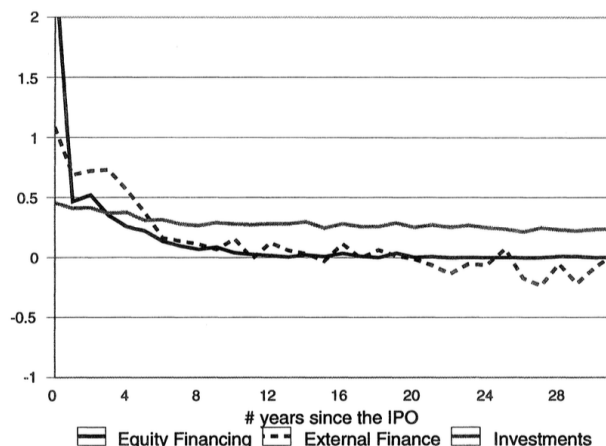


FIGURE 1. LIFE CYCLE OF EXTERNAL FINANCING AND INVESTMENTS

Notes: This graph plots the median level of external financing, equity financing, and investments in the United States across three-digit SIC industries as a function of the number of years since the IPO. External finance is the amount of capital expenditures not financed with cash flow from operations, reduction in inventories, or decreases in trade credit. Equity finance is the net amount of funds raised through equity issues divided by the amount of investments. Investment is the ratio of capital expenditures to net property, plant, and equipment. The IPO year is defined as the first year in which a company starts to be traded on the NYSE, AMEX, or NASDAQ. All the information is obtained from the flow-of-funds data in *Compustat*, except for the SIC code which is from the Center for Research on Securities Prices.

4.6 External-finance channel versus general cost of capital

The cash-flow and investment-intensity regressions show that the results are not simply about capital-intensive sectors doing better where capital is cheap. The paper argues that part of the effect operates specifically through a lower **incremental cost of external finance**.

Interpretation. This is central to the paper's claim: financial development is not just another name for a lower cost of capital; it changes how easily firms can fund investment when internal cash is insufficient.

5 Tables and figures

5.1 Figure 1 and Table 1: life cycle of financing and industry rankings

Read:

- Figure 1 shows a sharp life-cycle pattern. Around the IPO, firms rely heavily on **equity financing** and **external finance**; by roughly year 10, net external finance is close to zero.
- Investment intensity remains positive throughout the life cycle, but external finance is mostly an **early-firm phenomenon**.
- Table 1 ranks industries by dependence. At the top are **Drugs (1.49)**, **Plastic products (1.14)**, **Office and computing (1.06)**, and **Radio (1.04)**.
- At the bottom are **Tobacco (-0.45)**, **Pottery (-0.15)**, and **Leather (-0.14)**.

Main takeaway. The paper's external-dependence index has strong economic content: some industries systematically need outside finance, especially early in firm life.

TABLE 1—PATTERN OF EXTERNAL FINANCING AND INVESTMENT ACROSS INDUSTRIES
IN THE UNITED STATES DURING THE 1980'S

ISIC code	Industrial sectors	All companies		Mature companies		Young companies	
		External dependence	Capital expenditures	External dependence	Capital expenditures	External dependence	Capital expenditures
314	Tobacco	-0.45	0.23	-0.38	0.24	—	—
361	Pottery	-0.15	0.20	0.16	0.41	-0.41	0.13
323	Leather	-0.14	0.21	-1.33	0.27	-1.53	0.16
3211	Spinning	-0.09	0.16	-0.04	0.19	—	—
324	Footwear	-0.08	0.25	-0.57	0.23	0.65	0.26
372	Nonferrous metal	0.01	0.22	0.07	0.21	0.46	0.24
322	Apparel	0.03	0.31	-0.02	0.27	0.27	0.37
353	Petroleum refineries	0.04	0.22	-0.02	0.22	0.85	0.28
369	Nonmetal products	0.06	0.21	0.15	0.22	-0.03	0.26
313	Beverages	0.08	0.26	-0.15	0.28	0.63	0.26
371	Iron and steel	0.09	0.18	0.09	0.16	0.26	0.19
311	Food products	0.14	0.26	-0.05	0.25	0.66	0.33
3411	Pulp, paper	0.15	0.20	0.13	0.21	0.22	0.20
3513	Synthetic resins	0.16	0.30	-0.23	0.20	0.79	0.45
341	Paper and products	0.18	0.24	0.10	0.23	0.57	0.29
342	Printing and publishing	0.20	0.39	0.14	0.33	0.60	0.41
352	Other chemicals	0.22	0.31	-0.18	0.25	1.35	0.46
355	Rubber products	0.23	0.28	-0.12	0.21	0.50	0.32
332	Furniture	0.24	0.25	0.33	0.17	0.68	0.29
381	Metal products	0.24	0.29	0.04	0.25	0.87	0.34
3511	Basic excluding fertilizers	0.25	0.30	0.08	0.24	0.79	0.29
331	Wood products	0.28	0.26	0.25	0.23	0.34	0.40
384	Transportation equipment	0.31	0.31	0.16	0.28	0.58	0.31
354	Petroleum and coal products	0.33	0.23	0.16	0.26	-0.26	0.22
3843	Motor vehicle	0.39	0.32	0.11	0.33	0.76	0.32
321	Textile	0.40	0.25	0.14	0.24	0.66	0.26
382	Machinery	0.45	0.29	0.22	0.25	0.75	0.33
3841	Ship	0.46	0.43	0.04	0.34	1.05	0.56

TABLE 1—Continued.

ISIC code	Industrial sectors	All companies		Mature companies		Young companies	
		External dependence	Capital expenditures	External dependence	Capital expenditures	External dependence	Capital expenditures
390	Other industries	0.47	0.37	-0.05	0.28	0.80	0.49
362	Glass	0.53	0.28	0.03	0.28	1.52	0.33
383	Electric machinery	0.77	0.38	0.23	0.29	1.22	0.46
385	Professional goods	0.96	0.45	0.19	0.33	1.63	0.52
3832	Radio	1.04	0.42	0.39	0.30	1.35	0.48
3825	Office and computing	1.06	0.60	0.26	0.38	1.16	0.64
356	Plastic products	1.14	0.44	—	—	1.14	0.48
3522	Drugs	1.49	0.44	0.03	0.32	2.06	0.47

Notes: This table reports the median level of external financing and capital expenditure for ISIC industries during the 1980's. External dependence is the fraction of capital expenditures not financed with cash flow from operations. Cash flow from operations is broadly defined as the sum of *Compustat* funds from operations (item #110), decreases in inventories, decreases in receivables, and increases in payables. Capital expenditures are the ratio of capital expenditures to net property plant and equipment. Mature companies are firms that have been public for at least ten years; correspondingly, young companies are firms that went public less than ten years ago. The year of going public is the first year in which a company starts to be traded on the NYSE, AMEX, or NASDAQ. All companies is the union of mature and young firms plus firms for which the year of going public could not be determined (firms already traded on NASDAQ in 1972). All the information is obtained from the flow-of-funds data in *Compustat*, except for the SIC code which is obtained from the Center for Research on Securities Prices and then matched with the ISIC code.

5.2 Table 2 and Table 3: country finance measures and summary statistics

Read:

- Table 2 documents wide cross-country heterogeneity in financial development, from poorer economies with weak capitalization and disclosure to richer economies with much deeper financial systems.
- Table 3 shows that average annual real industry growth is about **3.4%**.
- Mean external dependence across industries is about **0.319**.
- Mean total capitalization/GDP is **0.738**, mean domestic credit/private sector is **0.377**, and mean accounting standards are **61.3**.
- The U.S. and Canadian dependence measures are strongly correlated (≈ 0.77), which supports the portability of the measure.

Main takeaway. There is plenty of variation in both the industry-side and country-side ingredients needed for identification.

5.3 Table 4: baseline interaction results

Read:

TABLE 2—FINANCIAL DEVELOPMENT ACROSS COUNTRIES

Country	Accounting standards	Total capitalization over GDP	Domestic credit to private sector over GDP	Per capita income (dollars)
Bangladesh	—	0.20	0.07	121
Kenya	—	0.28	0.20	417
Morocco	—	0.41	0.16	807
Sri Lanka	—	0.44	0.21	252
Pakistan	—	0.53	0.25	290
Costa Rica	—	0.53	0.26	2,155
Zimbabwe	—	1.01	0.30	441
Jordan	—	1.16	0.54	1,109
Egypt	24	0.74	0.21	563
Portugal	36	0.82	0.52	2,301
Peru	38	0.28	0.11	842
Venezuela	40	0.34	0.30	3,975
Colombia	50	0.21	0.14	1,150
Turkey	51	0.35	0.14	1,081
Chile	52	0.74	0.36	2,531
Brazil	54	0.33	0.23	1,650
Austria	54	1.00	0.77	9,554
Greece	55	0.74	0.44	3,814
India	57	0.50	0.24	240
Mexico	60	0.39	0.16	2,651
Belgium	61	0.65	0.29	11,226
Denmark	62	0.56	0.42	12,188
Germany	62	1.08	0.78	12,345
Italy	62	0.98	0.42	6,460
Korea	62	0.63	0.50	1,407
Netherlands	64	0.91	0.60	11,155
Spain	64	1.02	0.76	5,087
Israel	64	1.18	0.67	3,573
Philippines	65	0.46	0.28	729
Japan	65	1.31	0.86	9,912

TABLE 2—Continued.

Country	Accounting standards	Total capitalization over GDP	Domestic credit to private sector over GDP	Per capita income (dollars)
France	69	0.70	0.54	11,337
New Zealand	70	0.59	0.19	7,490
South Africa	70	1.51	0.26	2,899
Norway	74	0.63	0.34	13,430
Canada	74	0.98	0.45	10,486
Australia	75	0.82	0.28	9,866
Malaysia	76	1.19	0.48	1,683
Finland	77	0.52	0.48	10,181
U.K.	78	0.78	0.25	9,600
Singapore	78	1.96	0.57	4,661
Sweden	83	0.79	0.42	14,368

Notes: Accounting standards is an index developed by the Center for International Financial Analysis and Research ranking the amount of disclosure in annual company reports in each country. Total capitalization to GDP is the ratio of the sum of equity market capitalization (as reported by the IFC) and domestic credit (IFS lines 32a–32f but not 32e) to GDP. Domestic credit to the private sector is IFS line 32d. Per capita income in 1980 is in dollars and is from the IFS.

- Using **total capitalization**, the interaction coefficient is **0.069** with a differential growth effect of about **1.3 percentage points**.
- Using **domestic credit/private sector**, the interaction is **0.118**, with differential growth of about **1.1 points**.
- Using **accounting standards**, the interaction is **0.155**, with differential growth of about **0.9 points**.
- Using **instrumented accounting standards**, the interaction rises to about **0.165**, with differential growth around **1.0 point**.
- When accounting standards and capitalization are included together, the capitalization interaction becomes tiny, while the accounting-standards interaction remains strong.

Main takeaway. Industries that depend more on outside finance grow disproportionately faster in countries with more developed financial systems, and the disclosure-based measure appears especially informative.

5.4 Table 5 and Table 6: actual growth patterns and young-firm dependence

Read:

- Table 5 is a very intuitive reduced-form check. In countries **below** the median in accounting standards, the least dependent industries grow faster than the most dependent ones. In countries

TABLE 3—SUMMARY STATISTICS

A: Summary Statistics						
Variable	Mean	Median	Standard deviation	Minimum	Maximum	Number of observations
Industry's real growth	0.034	0.029	0.099	-0.447	1.000	1242
Industry's growth in number of firms	0.012	0.007	0.071	-0.414	0.759	1073
Industry's growth in average firms' size	0.022	0.026	0.094	-0.536	0.410	1070
Industry's share of total value added	0.016	0.009	0.021	0.000	0.224	1217
Log per capita income in 1980 in dollars	7.870	7.971	1.344	4.793	9.573	41
Average years of schooling	5.900	5.442	2.829	1.681	12.141	41
External finance dependence (all firms)	0.319	0.231	0.319	-0.451	1.492	36
External finance dependence (old firms)	0.010	0.075	0.302	-1.330	0.394	35
External finance dependence (young firms)	0.675	0.673	0.643	-1.535	2.058	34
External finance dependence (1970's)	0.078	0.073	0.188	-0.450	0.542	35
External finance dependence (Canadian firms)	0.427	0.384	0.767	-0.802	3.512	27
Cash flow generated	0.173	0.198	0.112	-0.217	0.331	36
Investment intensity	0.298	0.278	0.095	0.161	0.600	36
Total capitalization over GDP	0.738	0.696	0.375	0.199	1.962	41
Domestic credit to private sector over GDP	0.377	0.302	0.201	0.069	0.856	41
Accounting standards	61.324	62.000	13.238	24.000	83.000	34
Accounting standards (1983)	65.393	68.500	11.426	39.000	81.000	28

B: Correlation Between Measures of External Dependence

	All	Old	Young	1970's	Cash flow	Investment
External finance dependence (all firms)	1.00	—	—	—	—	—
External finance dependence (old firms)	0.46 (0.01)	1.00	—	—	—	—
External finance dependence (young firms)	0.72 (0.00)	0.48 (0.00)	1.00	—	—	—
External finance dependence (1970's)	0.63 (0.00)	0.42 (0.01)	0.48 (0.00)	1.00	—	—
Cash flow generated	-0.91 (0.00)	-0.37 (0.03)	-0.55 (0.00)	-0.50 (0.00)	1.00	—
Investment intensity	0.81 (0.00)	0.28 (0.10)	0.64 (0.00)	0.63 (0.00)	-0.60 (0.00)	1.00
External finance dependence (Canadian firms)	0.77 (0.00)	0.36 (0.07)	0.58 (0.00)	0.37 (0.07)	-0.78 (0.00)	0.55 (0.00)

TABLE 3—Continued.

C: Correlation Between Measures of Financial Development					
	Total capitalization	Market capitalization	Domestic credit to private sector	Accounting standards	Accounting standards 1983
Market capitalization over GDP	0.79 (0.00)	—	—	—	—
Domestic credit to private sector over GDP	0.67 (0.00)	0.21 (0.18)	1.00	—	—
Accounting standards	0.41 (0.02)	0.45 (0.01)	0.25 (0.17)	1.00	—
Accounting standards (1983)	0.27 (0.17)	0.39 (0.05)	-0.14 (0.50)	0.68 (0.00)	1.00
Per capita income	0.26 (0.09)	0.04 (0.80)	0.48 (0.00)	0.56 (0.00)	0.28 (0.16)

Notes: Industry real growth is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. The growth in the number of firms is the difference between the log of number of ending-period firms and the log of number of beginning-period firms. The average size of firms in the industry is obtained by dividing the value added in the industry by the number of firms, and the growth in average size is obtained again as a difference in logs. The industry's share of total value added is computed dividing the 1980 value added of the industry by the total value added in manufacturing that year. External dependence is the median fraction of capital expenditures not financed with cash flow from operations for each industry. Cash flow from operations is broadly defined as the sum of *Compustat* funds from operations (items #110), decreases in inventories, decreases in receivables, and increases in payables. External dependence has been constructed using *Compustat* firms between 1980 and 1990, except for Canada where we use *Global Vantage* (Standard & Poor's, 1993) data between 1982 and 1990. Accounting standards is an index developed by the Center for International Financial Analysis and Research ranking the amount of disclosure of companies' annual reports in each country. In Panels B and C the *p*-values are reported in parentheses.

TABLE 4—INDUSTRY GROWTH AND VARIOUS MEASURES OF DEVELOPMENT

Variable	Financial development measured as				
	Total capitalization	Bank debt	Accounting standards	Accounting standards in 1983	Accounting standards and capitalization
Industry's share of total value added in manufacturing in 1980	-0.912 (0.246)	-0.899 (0.245)	-0.643 (0.204)	-0.587 (0.223)	-0.443 (0.135)
Interaction (external dependence × total capitalization)	0.069 (0.023)	—	—	—	0.012 (0.014)
Interaction (external dependence × domestic credit to private sector)	—	0.118 (0.037)	—	—	—
Interaction (external dependence × accounting standards)	—	—	0.155 (0.034)	—	0.133 (0.034)
Interaction (external dependence × accounting standards 1983)	—	—	—	0.099 (0.036)	—
<i>R</i> ²	0.290	0.290	0.346	0.239	0.419
Number of observations	1217	1217	1067	855	1042
Differential in real growth rate	1.3	1.1	0.9	0.4	1.3

Notes: The dependent variable is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. External dependence is the fraction of capital expenditures not financed with internal funds for U.S. firms in the same industry between 1980–1990. The interaction variable is the product of external dependence and financial development. Financial development is total capitalization in the first column, domestic credit to the private sector over GDP in the second column, accounting standards in 1990 in the third column, and accounting standards in 1983 in the fourth column. The sixth column is estimated with instrumental variables. Both the coefficient estimate for the interaction term and the standard error when accounting standards is the measure of development are multiplied by 100. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

TABLE 5—EFFECT OF FINANCIAL DEVELOPMENT ON ACTUAL GROWTH RATES IN DIFFERENT INDUSTRIES

	Countries below the median in accounting standards	Countries above the median in accounting standards
Least financially dependent industries		
Tobacco	0.53	-0.60
Pottery	0.25	-0.30
Leather	0.77	-0.77
Most financially dependent industries		
Drug	-1.11	1.30
Plastics	-0.21	0.21
Computers	-2.00	1.80

Notes: This table reports the mean residual growth rate (in percentage terms) obtained after regressing the annual compounded growth rate in real value added for the period 1980–1990 on industry and country dummies.

above the median, the ranking reverses.

- For example, below-median-finance countries, **Drugs** have residual growth **-1.11** while **Tobacco** has **0.53**; above-median-finance countries, **Drugs** rise to **1.30** while **Tobacco** falls to **-0.60**.
- Table 6 repeats the baseline regressions using dependence measured only for **young firms**. The interaction remains positive and significant, but the implied growth differentials are smaller, around **0.4–0.6 points**.

Main takeaway. The core pattern is visible directly in the data, and the results do not rely on one particular way of measuring dependence.

5.5 Table 7 and Table 8: alternative dependence measures and decomposition of growth

TABLE 7—INDUSTRY GROWTH AND VARIOUS MEASURES OF EXTERNAL DEPENDENCE

Variable	External dependence measured using		
	Old firms	Firms in 1970's	Canadian firms
Industry's share of total value added in manufacturing in 1980	-0.625 (0.204)	-0.620 (0.205)	-0.610 (0.235)
Interaction (external dependence × accounting standards)	0.253 (0.063)	0.315 (0.127)	0.065 (0.023)
R ²	0.336	0.334	0.343
Number of observations	1035	1035	802
Differential in real growth rate	0.9	0.9	0.8

Notes: The dependent variable is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. External dependence is the fraction of capital expenditures not financed with internal funds by firms in the same industry during the 1980's. In the first column this ratio is computed only for companies that have been public for at least ten years. In the second column it is computed for U.S. firms during the 1970's. In the third column it is computed for Canadian firms during the 1980's. Also in the third column, data on U.S. industries are included while data on Canadian industries are dropped. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions are estimated using instrumental variables and include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

TABLE 6—INDUSTRY GROWTH AND VARIOUS MEASURES OF DEVELOPMENT USING EXTERNAL DEPENDENCE MEASURED FOR YOUNG FIRMS

Variable	Financial development measured as				
	Total capitalization	Bank debt	Accounting standards	Accounting standards in 1983	Accounting standards and capitalization
Industry's share of total value added in manufacturing in 1980	-0.911 (0.287)	-0.904 (0.286)	-0.568 (0.234)	-0.616 (0.252)	-0.293 (0.149)
Interaction (external dependence × total capitalization)	0.021 (0.012)	—	—	—	-0.004 (0.008)
Interaction (external dependence × domestic credit to private sector)	—	0.034 (0.019)	—	—	—
Interaction (external dependence × accounting standards)	—	—	0.046 (0.021)	—	0.045 (0.022)
Interaction (external dependence × accounting standards 1983)	—	—	—	0.038 (0.019)	—
R ²	0.283	0.283	0.341	0.236	0.415
Number of observations	1150	1150	1008	808	984
Differential in real growth rate	0.6	0.5	0.4	0.2	0.1

Notes: The dependent variable is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. External dependence is the fraction of capital expenditures not financed with internal funds between 1980–1990 for U.S. firms which went public in the previous ten years belonging to the same industry. The interaction variable is the product of external dependence and financial development. Financial development is total capitalization in the first column, domestic credit to the private sector over GDP in the second column, accounting standards in 1990 in the third column, and accounting standards in 1983 in the fourth column. The sixth column is estimated with instrumental variables. Both the coefficient estimate for the interaction term and the standard error when accounting standards is the measure of development are multiplied by 100. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

TABLE 8—GROWTH IN AVERAGE SIZE AND NUMBER OF ESTABLISHMENTS

Variable	External dependence measured using					
	All firms		Young firms		Mature firms	
	Growth average size	Growth number	Growth average size	Growth number	Growth average size	Growth number
Industry's share of total value added in manufacturing in 1980	-0.620 (0.217)	-0.312 (0.154)	-0.635 (0.256)	-0.252 (0.179)	-0.624 (0.220)	-0.282 (0.152)
Interaction (external dependence × accounting standards)	0.051 (0.043)	0.115 (0.037)	-0.021 (0.029)	0.078 (0.024)	0.125 (0.055)	0.131 (0.041)
R ²	0.498	0.314	0.500	0.302	0.492	0.310
Number of observations	951	975	899	922	923	947
Differential in real growth rate	0.3	0.7	-0.2	0.6	0.4	0.4

Notes: The average size of establishments in the industry is obtained by dividing the value added in the industry by the number of establishments, and the growth in average size is obtained as a difference in logs between average size in 1990 and average size in 1980. The growth in the number of establishments is the log of the number of establishments in 1990 less the log of the number of establishments in 1980. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions are estimated using instrumental variables and include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

Read:

- Table 7 shows robustness to using **mature U.S. firms**, **U.S. firms in the 1970s**, and **Canadian firms**. The interaction remains positive and significant, with differential growth rates around **0.8–0.9 points**.
- Table 8 decomposes industry growth. With dependence measured across all firms, the interaction is **0.051** for **growth in average size** but **0.115** for **growth in number of establishments**.
- The implied differential growth effect is about **0.3 points** for average size and **0.7 points** for the number of establishments.

Main takeaway. Financial development matters not only for incumbent expansion, but especially for the **creation of new establishments**.

5.6 Table 9 and Table 10: omitted variables, reverse causality, and the cost-of-capital channel

Variable	Human capital	Economic development	Above median	Below median
Industry's share of total value added in manufacturing in 1980	−0.386 (0.137)	−0.422 (0.134)	−0.437 (0.178)	−6.079 (1.932)
Interaction (external dependence × accounting standards)	0.191 (0.072)	0.149 (0.055)	0.161 (0.065)	0.161 (0.066)
Interaction 2 (external dependence × average years of schooling)	−0.002 (0.003)	—	—	—
Interaction 3 (external dependence × log of per capita income in 1980)	—	0.000 (0.005)	—	—
R^2	0.413	0.418	0.548	0.390
Number of observations	1006	1042	522	545
Differential in real growth rate	1.0	0.9	0.9	1.0

Notes: The dependent variable is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. The first column adds to the basic specification the interaction between external dependence and a country's human capital. The second column adds to the basic specification the interaction between external dependence and a country's level of economic development (log per capita income). The third column estimates the basic specification for industries that in 1980 were above the median industry in terms of the fraction they accounted for of value added in the manufacturing sector. The fourth column estimates the basic specification for industries that in 1980 were below the median industry in terms of the fraction they accounted for of value added in the manufacturing sector. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions are estimated using instrumental variables and include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

Variable	Cash flow intensity	Investment intensity	Both	Both measured for 1980
Industry's share of total value added in manufacturing in 1980	−0.588 (0.201)	−0.653 (0.205)	−0.639 (0.205)	−0.639 (0.207)
Interaction (internal cash flow × financial development)	0.482 (0.153)	—	−0.261 (0.196)	−0.595 (0.295)
Interaction 2 (investment intensity × accounting standards)	—	0.623 (0.221)	0.443 (0.283)	0.800 (0.299)
R^2	0.343	0.345	0.345	0.344
Number of observations	1067	1067	1067	1035
Differential in real growth rate	−0.7	1.4	0.5	1.6

Notes: The dependent variable is the annual compounded growth rate in real value added for the period 1980–1990 for each ISIC industry in each country. Internal cash flow is the ratio of cash flow from operations broadly defined (see text) to net property plant and equipment for U.S. firms in the same industry. Investment intensity is the ratio of capital expenditures to property plant and equipment for U.S. firms in the same industry. The fourth column uses the cash flow intensity and the investment intensity measured for the year 1980. The differential in real growth rate measures (in percentage terms) how much faster an industry at the 75th percentile level of external dependence grows with respect to an industry at the 25th percentile level when it is located in a country at the 75th percentile of financial development rather than in one at the 25th percentile. All regressions are estimated using instrumental variables and include both country and industry fixed effects (coefficient estimates not reported). Heteroskedasticity robust standard errors are reported in parentheses.

Read:

- Table 9 shows the main interaction survives controls for **human capital** and **per capita income**. The coefficient stays around **0.149–0.191**.
- The result is also stable when the sample is restricted to industries **above the median** or **below the median** of a country's manufacturing sector in 1980.
- Table 10 separates **cash flow intensity** from **investment intensity**. When both are entered using 1980 measures, both matter, suggesting the result is not simply a generic cost-of-capital story.

Main takeaway. The finance-growth interaction is hard to dismiss as a proxy for human capital, general development, industry maturity, reverse causality, or simple capital intensity.

6 Short summary

- **Method.** Interact a country's financial development with an industry's technological dependence on external finance, while absorbing country and industry effects.

- **Identification.** Cross-industry differences within countries provide the key variation; legal origin and legal-system quality instrument for accounting standards.
- **Application.** Manufacturing industries across 41 countries over 1980–1990, with external dependence measured from U.S. Compustat firms.
- **Main result.** Financially dependent industries grow significantly faster in financially developed countries, especially through the growth in the **number of establishments**.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides evidence that financial development promotes growth through a specific channel: it reduces the cost of external finance to firms.
- This effect is strongest in industries that technologically rely more heavily on outside funds.
- The mechanism seems particularly important for **new establishment formation**, not just incumbent expansion.

7.2 Economic intuition

- Some industries can finance investment internally; others cannot.
- In financially underdeveloped countries, those external-finance-dependent industries are held back.
- Better accounting, deeper markets, and stronger financial intermediation allow these industries to fund investment and scale up.
- Because many new ideas may come from young or newly established firms, the long-run effect of finance can be larger than what a static investment view would suggest.

7.3 Takeaway

This paper is important because it turns a broad finance-and-growth correlation into a much sharper empirical test. Its central lesson is that **financial development is a source of comparative advantage for industries that need external funds**. Finance does not just accompany growth; it helps determine **which sectors grow, where new establishments appear, and how countries specialize**.